

Rento

Requirements Analysis Document

Rental Property Platform | Egypt-Japan University of Science and Technology

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Project Overview

Rento is a web-based rental property platform developed to address the real challenges faced by apartment seekers in Egypt, particularly the prevalence of misleading broker information. The platform enables property owners to list apartments with accurate details and photos, while renters can search, explore on an interactive map, save favorites, and book properties — all within a secure, authenticated environment. The system is built with HTML, CSS, and JavaScript, using a multi-page architecture with shared authentication and a persistent dark-mode preference.

1. Functional Requirements

Functional requirements define what the Rento system must do — the specific behaviours, features, and operations it must support to meet user needs.

1.1 User Authentication & Account Management

ID	Requirement	Description
FR-AUTH-01	User Registration	New users can create an account by providing name, email, phone number, and a password. Passwords must be confirmed and must be at least 6 characters long.
FR-AUTH-02	User Login	Registered users can log in with their email and password. A 'Remember Me' option persists the email field across sessions.
FR-AUTH-03	Google OAuth Login	Users may authenticate using their Google account via OAuth, bypassing manual registration.
FR-AUTH-04	Session Persistence	Upon successful login or registration, a user session is stored in localStorage so the user remains logged in across page navigations.
FR-AUTH-05	Logout	Authenticated users can log out from any page via the user dropdown in the navbar. Logging out clears the session and redirects to the home page.
FR-AUTH-06	Redirect if Already Logged In	If a user visits the Login or Register page while already authenticated, they are automatically redirected to the Explore (Map) page.

1.2 Navigation & Access Control

ID	Requirement	Description
FR-NAV-01	Auth-Gated Navigation	The nav links for Explore, Dashboard, Favorites, and My Rentals are restricted. Unauthenticated users clicking these links see a modal prompting login or registration.
FR-NAV-02	Public Pages	Home and About pages are accessible to all users without authentication.
FR-NAV-03	Dynamic Navbar State	When logged out, the navbar displays Login and Register buttons. When logged in, these are replaced by a user avatar button showing the user's initial and first name.
FR-NAV-04	User Dropdown Menu	Clicking the avatar button reveals a dropdown with quick links to Dashboard, My Rentals, Favorites, and a Logout button, plus the user's full name and email.
FR-NAV-05	Responsive Mobile Menu	On screens narrower than 900 px, the desktop navbar collapses to a hamburger icon that opens a slide-down mobile menu with all navigation links and auth state.
FR-NAV-06	Active Page Indicator	The currently active page link is visually highlighted in the navbar on every page.

1.3 Property Browsing & Search

ID	Requirement	Description
FR-PROP-01	Property Listing on Home Page	The home page displays rental property cards showing a photo, star rating, city, region, monthly price (in L.E.), and availability status (Available / Unavailable).
FR-PROP-02	Load More Properties	Authenticated users can trigger loading of additional property listings via a 'Load More Places' control at the bottom of the listing section.
FR-PROP-03	Interactive Map Search	The Explore page embeds a Leaflet.js map. Users can browse properties geographically, with markers indicating listing locations.
FR-PROP-04	Filter Sidebar	The Explore page includes a filter sidebar allowing users to narrow results by city, price range, property type, and other attributes. A 'Reset All' button clears all filters.
FR-PROP-05	Property Detail View	Selecting a property opens the Details page, which shows an image slider, full property specifications, pricing, and booking options.
FR-PROP-06	Image Slider	The Details page features a multi-image slider with previous/next navigation controls to showcase multiple photos of the property.

1.4 Favorites Management

ID	Requirement	Description
FR-FAV-01	Toggle Favorite	Authenticated users can mark or unmark any property as a favorite using a heart button on the property card. The icon updates immediately to reflect the state.
FR-FAV-02	Favorites Page	The Favorites page lists all properties saved by the authenticated user.
FR-FAV-03	Filter Favorites by City	Users can filter their saved favorites by city (All, Cairo, Alexandria, Other Cities).
FR-FAV-04	Sort Favorites	Users can sort their saved properties by criteria such as price or rating.

1.5 Booking

ID	Requirement	Description
FR-BOOK-01	Booking Form	Authenticated users can submit a booking request by filling in name, email, phone, move-in date, and duration on the Booking page.
FR-BOOK-02	Booking Confirmation	On successful form submission the system acknowledges the booking request.
FR-BOOK-03	Auth Guard on Booking	'More Details' and 'Choose Your Place' buttons on the home page redirect unauthenticated users to a login prompt rather than the booking flow.

1.6 Dashboard & My Rentals

ID	Requirement	Description
FR-DASH-01	Dashboard Overview	The Dashboard page presents a personalised welcome banner with the user's name, followed by summary statistics cards (active rentals, total spent, saved properties, etc.).
FR-DASH-02	My Rentals — Tab Navigation	The My Rentals page organises rental agreements into three tabs: Active, Upcoming, and Past, each showing the count of agreements in that category.
FR-DASH-03	Rental Card Details	Each rental card shows property name, address, dates, monthly price, and status badge.

2. Non-Functional Requirements

Non-functional requirements specify the quality attributes of the Rento system — how well it performs its functions. Usability requirements are given special emphasis as they directly affect the user experience and adoption of the platform.

2.1 Usability Requirements

Usability is the primary quality concern for Rento. The platform targets a broad Egyptian audience including first-time apartment seekers who may not be highly tech-savvy. The following usability requirements have been identified:

ID	Category	Requirement	Metric / Target
NFR-USA-01	Learnability	Intuitive Navigation Structure	A first-time user must be able to understand the main navigation and reach any core feature within 3 clicks without consulting a help guide.
NFR-USA-02	Feedback & Visibility	Clear Auth State Communication	The navbar must clearly indicate whether the user is logged in or out at all times. Login/Register buttons must disappear immediately on authentication.

ID	Category	Requirement	Metric / Target
NFR-USA-03	Error Prevention	Form Validation	All input forms (login, register, booking) must validate data client-side before submission and display inline, human-readable error messages rather than browser defaults.
NFR-USA-04	Consistency	Uniform Component Behaviour	All pages must use the same navbar, button styles, typography, and colour tokens. A shared CSS variable system enforces visual consistency.
NFR-USA-05	Accessibility — Colour Contrast	WCAG AA Contrast Compliance	Text on all backgrounds must achieve a contrast ratio of at least 4.5:1 (normal text) and 3:1 (large text) in both light and dark modes.
NFR-USA-06	Dark Mode Support	System-Wide Dark Theme	Every page must support a toggleable dark mode. The preference must persist via localStorage and be applied before the first paint to avoid a white flash.
NFR-USA-07	Mobile Friendliness	Responsive Design	All pages must be fully usable on screen widths from 320 px to 1920 px. The navbar must collapse gracefully below 900 px with a functional hamburger menu.
NFR-USA-08	Discoverability	Auth Modal on Restricted Access	When an unauthenticated user attempts to access restricted content, a contextual modal must appear explaining why access is restricted and offering clear Login/Register calls to action.
NFR-USA-09	Efficiency	Minimal Steps to Core Actions	A returning user must be able to log in and reach a property listing in no more than 4 user actions.

ID	Category	Requirement	Metric / Target
NFR-USA-10	Help & Documentation	Inline Contextual Cues	Placeholder text, label text, and button labels must be descriptive enough that users do not need external documentation to complete standard tasks.

2.2 Performance Requirements

ID	Category	Requirement	Metric / Target
NFR-PERF-01	Performance	Page Load Time	Each page must fully render within 3 seconds on a standard broadband connection (≥ 10 Mbps).
NFR-PERF-02	Performance	Map Rendering	The interactive Leaflet map on the Explore page must be interactive (tiles loaded, markers visible) within 4 seconds.
NFR-PERF-03	Performance	Dark Mode Toggle Latency	Switching between light and dark mode must complete in under 300 ms with no visible layout shift.

2.3 Security Requirements

ID	Category	Requirement	Metric / Target
NFR-SEC-01	Security	Session Storage Safety	User session data stored in localStorage must not include plaintext passwords; only non-sensitive identity data (name, email, timestamp) may be persisted client-side.
NFR-SEC-02	Security	Route Protection	All pages requiring authentication must re-verify session state on every load, preventing direct URL access by unauthenticated users.

ID	Category	Requirement	Metric / Target
NFR-SEC-03	Security	Input Sanitisation	Form inputs must be sanitised before any processing to prevent XSS attacks.

2.4 Reliability & Maintainability

ID	Category	Requirement	Metric / Target
NFR-REL-01	Reliability	Graceful Degradation	If a property image fails to load, a styled placeholder must be displayed so the card layout does not break.
NFR-MAINT-01	Maintainability	Shared Codebase	Navbar HTML, auth logic, and dark-mode CSS must be maintained in single shared files (auth.js, dark.css) rather than duplicated across pages.
NFR-MAINT-02	Maintainability	CSS Variable System	All colours and spacing tokens must be defined as CSS custom properties in :root and [data-theme='dark'] so theme changes require editing only one location.
NFR-MAINT-03	Portability	Cross-Browser Compatibility	The platform must function correctly on the latest stable versions of Chrome, Firefox, Safari, and Edge without browser-specific hacks.

3. Requirements Summary

Category	Count	Key IDs
Functional — Authentication	6	FR-AUTH-01 to FR-AUTH-06
Functional — Navigation & Access Control	6	FR-NAV-01 to FR-NAV-06
Functional — Property Browsing & Search	6	FR-PROP-01 to FR-PROP-06

Functional — Favorites Management	4	FR-FAV-01 to FR-FAV-04
Functional — Booking	3	FR-BOOK-01 to FR-BOOK-03
Functional — Dashboard & My Rentals	3	FR-DASH-01 to FR-DASH-03
Total Functional Requirements	28	
Non-Functional — Usability	10	NFR-USA-01 to NFR-USA-10
Non-Functional — Performance	3	NFR-PERF-01 to NFR-PERF-03
Non-Functional — Security	3	NFR-SEC-01 to NFR-SEC-03
Non-Functional — Reliability & Maintainability	4	NFR-REL-01, NFR-MAINT-01 to NFR-MAINT-03
Total Non-Functional Requirements	20	
GRAND TOTAL	48	